Worksheet 1-6 & 1-7 B Algebraic Limits #2

AP Calculus Name_____

Directions: Evaluate the following limits, if possible. Show all work and use proper notation.

1. $\lim_{x\to 3} \frac{x^2+4x-21}{x+3}$

2. $\lim_{x\to -2} \frac{x^2+5x+6}{x^2-4}$

3. $\lim_{x\to 0} \frac{3x^5-7x^4}{4x^4+3x^8}$

4. $\lim_{x\to 5} \frac{x^2-2x-15}{x^2-10x+25}$

5. $\lim_{x\to 0} \frac{x^3+x^2+3x}{x^5+7x}$

6. $\lim_{x\to \pi} \frac{x^2+4}{3x}$

7. $\lim_{x\to 3} \frac{\ln(2x-5)}{4-2x}$

8. $\lim_{x\to -4} \frac{16-x^2}{x^2-3x-28}$

9. $\lim_{x\to 2} \frac{4x-8}{x^2-9x+14}$

10. $\lim_{x\to 0} 5e^2$

11. $\lim_{x\to \pi} \frac{x-\pi}{3x-3\pi}$

12. $\lim_{x\to \pi/3} \frac{\cos(3x)}{x}$

13. $\lim_{x \to 0} \dfrac{\dfrac{1}{x+3} - \dfrac{1}{3}}{x}$

14. $\lim_{x \to 7} \dfrac{\sqrt{x+9} - 4}{x - 7}$

15. $\lim_{x \to 0} \dfrac{x}{\dfrac{2}{x+5} - \dfrac{2}{5}}$

16. $\lim_{x \to 1} \dfrac{\dfrac{1}{3} - \dfrac{1}{3x}}{x - 1}$

17. $\lim_{x \to 0} \dfrac{\sqrt{2x+11} - \sqrt{11}}{x}$

18. $\lim_{x \to 0} \dfrac{\dfrac{1}{(x+2)^2} - \dfrac{1}{4}}{x}$

19. $\lim_{x \to 1} \dfrac{4}{x^2 + x - 2}$

20. $\lim_{x \to 5} \dfrac{x}{(x-5)^2}$